

Sensor Platform

IG Handle is the physical acquisition boundary for GRANDE. It records what is installed, how each device is reached, which frame and topic it owns, how time is stamped, and whether the device is currently available.

The deployed inventory is defined by `sensor_contract.yaml`, not by launch-file names. `sensor_network.yaml` records runtime endpoints and interface roles. `sensor_frames.yaml` records configured static transforms and their evidence state, while `sensor_models.yaml` records capabilities independent of one deployed serial.

Changing a physical transform requires measurement evidence. Runtime code may select a configured device but cannot patch its geometry or identity.

The host uses separate local and boat-facing network roles. Stable device identity uses serials, USB attributes, udev aliases, and explicit network endpoints. A changing `/dev/ttyACM*` number or switch port is not itself a new identity. The runtime network contract keeps separate DT100 and Ping360 endpoints on the sensor subnet. Both sonar device addresses are unconfigured; the host-side address is `192.168.2.10`. The checked-in `01-netplan.yaml` template uses the same `192.168.2.0/24` subnet. The networkd template retains the camera, LiDAR, sonar, and boat-facing addresses with `ignore-carrier: true`; Netplan 0.104 or later supports this setting. The protected deployed file is `/etc/netplan/99-heron-sensor-net.yaml`. Changes preserve its other settings and the disabled legacy `sensor-net.service`, which uses an obsolete boat address. Assigned host addresses do not establish sonar endpoint identity or commissioned wiring. The Heron-facing host address is `192.168.131.10`, matching the deployed networkd configuration; the base computer remains `192.168.131.1`.

Camera serial files reserve one calibration record per physical camera. The current records are explicitly uncalibrated placeholders and must not be treated as metric camera calibration. The sensor contract assigns ordered inspection roles so DEFECTOR and ORACLE consume role, topic, frame, and calibration state without product-name coupling.

Device stamps are preserved when available. IG Handle publishes timing inputs for downstream estimation and records raw sensor evidence without turning acquisition health into mission or actuator authority. GRANDE normally owns integrated recording. Passive IGHandle captures use the same persistent physical graph and own only their recorder; ending a capture preserves acquisition, DLiO, native mapping and the dashboard. Configuration changes remain deliberate maintenance operations, independent of recording.

Heron hardware-client deployment

Heron retains its existing Kinetic `robot_upstart` job and joins only IGHandle's physical master. The hardware composition retains rosserial, GPS and status functions while removing stock localization, model transforms and command publishers. MCU-owned RC remains a separate physical path; software inspection cannot establish its operation during ROS loss.

From IGHandle's active install, copy the existing deployment helper, hardware launch and shared client module to Heron over the direct boat-facing LAN:

```
source ~/catkin_ws/heron_ws/install-native/setup.bash
sensor_package=$(rospack find ig_handle)
master_module=$(python3 -c 'import sensors.ros_master; print(sensors.ros_master.__file__)')
scp -o ProxyCommand=none -o ProxyJump=none \
  "$sensor_package/config/network/deploy_heron_ighandle_client.sh" \
  "$sensor_package/config/network/heron_ighandle_client.launch" \
  "$master_module" administrator@192.168.131.1:/tmp/
```

On Heron, use its authorized administrator session:

```
sudo bash /tmp/deploy_heron_ighandle_client.sh install \  
  /tmp/heron_ighandle_client.launch /tmp/ros_master.py \  
  http://192.168.131.10:11311 192.168.131.1 br0
```

The helper prepares all replacements before touching the deployed job. It preserves the existing workspace setup, log configuration and privilege prefix; an unsupported launch/PID/wait shape leaves the job untouched. It installs `base.launch`, `ros-start`, the shared master-client module and one `ros.service` override. The unique backup under `/var/backups/grande-heron-ros/` preserves the prior four managed files and records their original absence where applicable. No service is restarted by installation or rollback.

The client waits for the fixed master before launching hardware. Master loss, changed PID or changed `/run_id` ends that client; the existing `ros.service` restarts its cgroup with fresh registrations. It creates no master and does not signal another service's processes. A deliberate service stop remains stopped. Only a separately approved stationary case may restart the actual boat job.

Rollback uses the backup path printed by installation:

```
sudo bash /tmp/deploy_heron_ighandle_client.sh rollback \  
  /var/backups/grande-heron-ros/<saved-backup>
```

After rollback, review the restored files before deliberately restarting the job. A failed restoration reports the retained backup rather than claiming success. Actual launcher compatibility, authorized Heron sudo access, cold boot, master replacement and RC independence remain unverified until the boat is reachable and the corresponding cases are approved.

IMU Ownership and Recovery

GRANDE's canonical physical IMU is the serial-qualified Xsens MTi-30 on IG Handle. Sensor contract ID 1 binds serial `0368319D` to `/dev/sensors/imu`, publishes only below `/sensors/imu`, and is owned by exactly one persistent Xsens service. That service joins IGHandle's sole physical master at `http://192.168.131.10:11311`; both the master and advertised host address come from the canonical network contract. The provider waits without opening another device when its selected master or exact serial is unavailable, restarts after USB loss, stale output, driver exit, or master replacement, and refuses duplicate publishers. It records immutable process identities and revalidates current ancestry before signaling anything. `sensor_bringup` observes this external owner for readiness but never opens or stops its serial port.

Internally owned providers are launched only when the ROS publisher graph is available and the required topics are unclaimed. The supervisor verifies each publisher against the current managed process tree before reporting it alive, uses bounded ROS XML-RPC calls, and signals only processes whose PID, immutable start time, and current ancestry are all revalidated. Publishers on disabled sensor topics remain untouched but are exposed explicitly in health output. Every `sensor_bringup` instance also holds the same advisory lease at `/run/lock/ig-handle-sensor-bringup.lock` for its full lifetime. Because both service and deliberate maintenance launches pass through this process, cameras and LiDAR cannot be started twice. The kernel releases the lease on exit or crash; stale file contents never establish ownership.

Fixed ROS Graph Profiles

The active deployment consists of `ig-handle-roscore-user.service`, `ig-handle-xsens-user.service`, and `ig-handle-sensor-bringup-user.service`. IGHandle owns the physical master even when Heron is disconnected. The dependent services bind their lifecycle to that master, use the same sensor leases, publisher checks, and serial identity checks, and restart only their owned processes after a master replacement. There is no automatic master switch or second physical graph. Core and sensor-launch services restart after unexpected clean exits as well as failures. Dependent install sections add master `.wants/` links when enabled, ensuring acquisition and GRANDE return after the master restarts. Explicit `systemctl --user stop` suppresses the unit's automatic restart. A later master restart or boot starts enabled dependent clients again.

User services source `GRANDE_WORKSPACE_SETUP` from the optional `~/.config/grande/environment` deployment configuration, with the conventional workspace below the current user's home as default. Network addresses come from `sensor_network.yaml`, accessed through `network_config.py`. The configured boat-facing address must remain assigned locally even when Heron is absent.

The stock Heron `/imu/*` and `/cv5/ros_mscl_node` surfaces describe an optional onboard MicroStrain installation. They are not aliases for the IG Handle Xsens. No Xsens topic is bridged into `/imu/data_raw`, and Xsens recovery grants no navigation or actuation authority. A Heron without the optional onboard sensor may retain inactive stock diagnostics, but GRANDE readiness uses the canonical `/sensors/imu/data` contract.

The Teensy timing design and its disabled-by-default electrical gates are specified in `sensor_timing.md`. Host camera, LiDAR, and IMU acquisition remains continuous; the firmware's trigger scheduler is a future commissioning surface rather than an enabled runtime claim.

Battery Identity and Telemetry

The platform inventory distinguishes one fixed compute battery from two interchangeable propulsion batteries. `IGHANDLE-01` is commissioned against the complete live JK BMS identity: BLE address, device name, model, hardware and software versions, serial number, and manufacturing date. Voltage, current, and charge behavior are measurements and never serve as identity. A host-wide lock prevents concurrent processes from owning the same BLE device, while periodic device-information queries re-establish that identity during continued telemetry acquisition. The `/ighandle_jk_bms` node publishes standard state on `/sense_ighandle`; its paired `/sense_ighandle/details` record carries identity, raw pack and cell fields, status, alarms, counters, and provenance with the same timestamp. Fleet admission requires both synchronized messages to retain that exact caller identity; another publisher cannot establish compute-pack identity.

`HERON-01` and `HERON-02` represent physical propulsion packs that the Heron MCU cannot identify electrically. They remain uncommissioned until the matching durable labels are physically present and inspected. After commissioning, GRANDE binds a labelled installation to physical `/sense_heron` observations through an append-only event ledger. A new logger session, publisher restart, sequence or timestamp rollback, or source-progress gap prevents an older selection from carrying forward without explicit reconfirmation. The resulting effective assignment is published on `/battery/heron_identity` and recorded beside the measurement streams. This separation allows both removable packs to build independent histories while keeping unassigned observations visible as `UNKNOWN`.

The installation ledger attributes selection and removal to the local `username@hostname` caller by default. A supplied `selected_by` value is retained as operator-entered provenance, not as authenticated user identity.

Persistent selection state, installation history, normalized samples, and figures default to `~/.local/share/grande/battery`. Boot infrastructure publishes compute-pack state on IGHandle's physical graph but does not start normalized battery logging. Deliberately started compute-only and combined loggers use the same graph and data root and are mutually exclusive. Repository data remains retained historical evidence rather than live mutable state.

Energy integration is available only from contiguous, identity-qualified compute pack power. Propulsion motor-controller currents are preserved for diagnosis but do not establish total pack current; propulsion energy therefore remains null with an explicit unavailability reason.

Independent Sonar Devices

IG Handle owns two separately selectable sensor identities. ID 8 is the DT100 echosounder under `/sensors/sonar/echosounder/`; ID 9 is the Ping360 imaging sonar under `/sensors/sonar/imaging/`. Each has a separate network endpoint, configured frame, provider process, and lifecycle state. Selecting neither, either one, or both uses the same sensor lifecycle owner. The shared Ethernet switch and arm cable are a proposed physical topology, not installed or electrically qualified by this software change. The box would require independently regulated power for the switch and each sonar. Its cable current, voltage drop, connectors, and installed device revisions still require design verification. The published DT100 supply range is 22–32 V; the Ping360 Ethernet wiring guide states 11–18 V.

ID 8 remains an uncommissioned passive UDP receiver. Its source address is blank and `hardware_commissioned` is false. Receiving an unverified datagram does not assign DT100 identity or make a decoded cloud. Once commissioned, the DT100 vendor output path admits only the configured source endpoint; MARINER's existing decoder accepts the documented 83P profile format and rejects incompatible or malformed formats. Its downstream products are raw packets, a decoded cloud, an accumulated cloud, and the latest slice. The selected 83P profile settings remain in the dedicated sonar profile file; they are admission settings, not evidence of physical configuration.

ID 9 uses the existing Ping protocol over UDP. The default mode requests identity only; it cannot produce scan imagery and must report no recent measurement. Active acoustic scanning requires both scan mode and the configured transmit allowance. In scan mode, ten seconds without a valid profile causes a one-time motor-off request and stops further scan requests. That UDP request has no device acknowledgement here; physical motor state must be checked separately. Profiles, a polar intensity image, raw packets, and diagnostics are kept distinct. Intensity is not geometric range.

The sensor supervisor reports provider process operation separately from measurement availability. Absent profiles in identity mode and old latched diagnostics are observations, not reasons to restart the provider. The dashboard displays both devices separately; DT100 uses the cloud viewer, and Ping360 uses the on-demand image viewer. A positive source-data label also requires verified publisher ownership. GRANDE's runtime and raw bag profiles include both configured sonar streams; narrower recording profiles have their own topic lists. MARINER can start both processing paths together. Optional Range Aid marker processing launches one independent frontend per enabled sonar. DT100 and Ping360 retain separate timestamps, extrinsic revisions, observations, statuses, and diagnostic visualizations. Both can run together without making either stream authoritative for navigation.

Configured frames `dt100_link` and `ping360_link` retain their seed extrinsic revisions, not measured calibration. DT100's down-looking fan and Ping360's horizontal sequential scan do not share a geometric model. The simulator can supply explicitly synthetic packets or profiles through the same downstream processing; simulation is not proof of physical propagation, endpoint identity, or calibration. Physical source, timing, frame and profile evidence must be retained before claiming a calibrated map or marker observation.

Heron Propulsion

IG Handle owns the physical facts needed to inspect and commission the Heron's propulsion hardware. MARINER owns controller mathematics and the final guarded command path; GRANDE owns evidence lineage and run-level evaluation.

The hardware chain includes battery protection, power distribution, vehicle controller, motor-enable/heartbeat path, Castle ESCs, motors, impellers, ducts, wiring, and telemetry sensing. Reverse asymmetry may come from configuration, electrical delivery, motor/ESC condition, mechanics, water loading, weight distribution, or command processing; one observation does not identify cause.

Restrained-air current can isolate electrical or mechanical asymmetry from water flow, but it does not identify afloat thrust. Current is not treated as force without an independent force measurement.

Castle Link can read and compare ESC profiles, including direction, endpoints, braking, timing, and current-related settings. The two sides should be compared as exact exported profiles before changing parameters. Physical inspection covers connectors, motor/shaft freedom, impeller and duct condition, water ingress, and side-to-side wiring differences.

ROS controller gains affect software command response but do not correct an ESC, motor, impeller, or hull-balance defect. Hardware and software changes are isolated and recorded separately.

Useful evidence aligns requested command, accepted drive, motor-enable state, voltage, side-specific current, RPM or PWM when available, and vehicle motion. Forward/reverse, left/right, coast, dry, restrained, and afloat regimes remain labelled. Replacement screening is a candidate record, not compatibility approval.